

NAME:
NEPTUN:

First Solution: **A**
Second Solution: **A**

1. A company's cash position, measured in millions of dollars, follows a generalized Wiener process with a drift rate of 5 per year and a variance rate of 8 per year. The current position is 50. What is the expected value, and the standard deviation of the cash position after two years?

- a) $EV = 60$; $STD = 4$
- b) $EV = 60$; $STD = 16$
- c) $EV = 10$; $STD = 16$
- d) $EV = 10$; $STD = 4$

$$EV = 50 + 2 \cdot 0,5 = 60$$

$$STD = \sqrt{8 \cdot 2} = 4$$

2. $N(x)$ is the standard normal distribution function. Calculate $N(0,129)$ if we know $N(-0,129) = 0,449$.

- a) $N(0,129) = 0,551$
- b) $N(0,129) = -0,551$
- c) $N(0,129) = -0,449$
- d) $N(0,129) = 1,449$

$$N(0,129) + N(-0,129) = 1$$

$$N(0,129) = 1 - 0,449 = 0,551$$